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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
iris = load_iris()
X = iris.data
y = iris.target

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, random_state=91
)

k = 5
knn = KNeighborsClassifier(n_neighbors=k)
knn.fit(X_train, y_train)

y_pred = []

for i in range(len(X_test)):
    pred = knn.predict([X_test[i]])
    y_pred.append(pred[0])

print("Index\tActual\t\tPredicted")
print("-----")

correct = 0

for i in range(len(y_test)):
    actual_label = iris.target_names[y_test[i]]
    predicted_label = iris.target_names[y_pred[i]]

    print(f"{i}\t{actual_label}\t{predicted_label}")

    if y_test[i] == y_pred[i]:
        correct += 1

manual_accuracy = correct / len(y_test)
print("\nManual Accuracy:", manual_accuracy)

print("\nSklearn Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n",
      classification_report(y_test, y_pred, target_names=iris.target_names))

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Index	Actual	Predicted
0	virginica	virginica
1	versicolor	versicolor
2	versicolor	versicolor
3	virginica	virginica
4	virginica	virginica
5	versicolor	versicolor
6	versicolor	versicolor
7	setosa setosa	
8	versicolor	versicolor
9	virginica	virginica
10	setosa setosa	
11	virginica	virginica
12	versicolor	versicolor
13	virginica	virginica
14	setosa setosa	
15	versicolor	versicolor
16	versicolor	versicolor
17	virginica	virginica
18	virginica	virginica

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19  setosa setosa
20  virginica virginica
21  versicolor virginica
22  setosa setosa
23  versicolor versicolor
24  versicolor versicolor
25  setosa setosa
26  setosa setosa
27  virginica virginica
28  setosa setosa
29  setosa setosa
30  versicolor versicolor
31  virginica virginica
32  versicolor versicolor
33  virginica virginica
34  setosa setosa
35  setosa setosa
36  versicolor versicolor
37  versicolor versicolor
38  setosa setosa
39  versicolor versicolor
40  virginica virginica
41  virginica virginica
42  virginica virginica
43  virginica virginica
44  versicolor versicolor
```

Manual Accuracy: 0.9777777777777777

Sklearn Accuracy: 0.9777777777777777

Confusion Matrix:

```
[[12  0  0]
 [ 0 16  1]
 [ 0  0 16]]
```

Classification Report: